ENS3553
Signals and Systems
Module 5
The Fourier Series

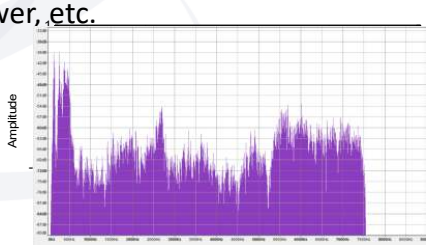


Overview

- This lecture will introduce the following topics:
  - Introduction to frequency response of CT systems
  - Representation of CT periodic signals in the frequency domain using the Fourier series
  - Identifying the characteristics of a frequency domain signal from its frequency domain representation
  - Amplitude and phase spectra of periodic signals and how they relate to the bandwidth and power of a signal

Frequency Domain Representation

- So far - **signals** viewed as (primarily) time-waveforms
  - i.e. functions of time, or **time-domain**
  - Provides some insights into the properties of the signal
- Representing signals in the **frequency-domain**
  - can gain many other useful insights
  - relating to spectral distribution, bandwidth, power, etc.
- How to determine the frequency components of a signal?



Frequency Response of Systems

- Previously, characterised a linear, time-invariant (LTI) **system** by its unit impulse response:

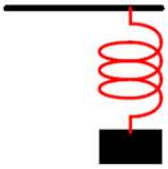
$$\text{CT}: y(t) = x(t) * h(t) = \int_{-\infty}^{\infty} x(\tau)h(t-\tau)d\tau$$

$$\text{DT}: y[n] = x[n] * h[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k]$$

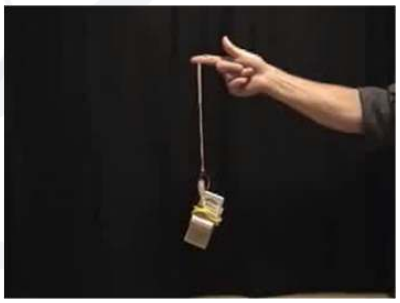
- Especially insightful for some systems
- Now, a different way to characterise a system: the **frequency-response**



Frequency Response Example

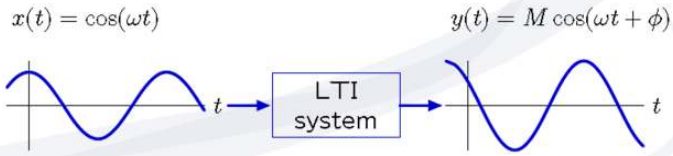


- Consider simple mass spring system
  - mass suspended from a spring attached to a rod that can move vertically
- What happens if the rod is moved up and down slowly?
- What happens as the oscillation rate is increased?
- What happens at very high oscillation rates (high frequencies)?
- This describes the frequency response of this system
- How can it be determined analytically?



LTI System Response to Periodic Signals

- What is the response of an LTI system to a periodic signal?
  - It is a periodic signal of the same frequency
- If input to an LTI is a sinusoid ⇒ output is a sinusoid with same frequency
  - possibly different amplitude
  - possibly different phase angle



- The frequency response: plot of the magnitude, M, and phase angle φ as a function of frequency, ω (or f)

Response of an LTI System to e^{st}

- Special class of signals: the everlasting exponential function e^{st}
  - where *s* is a complex variable
- An LTI system's (zero-state) response to an everlasting exponential input
  - is a scalar multiple of that input
  - no other input function has this property
- An **input** for which the *output is of the same (scaled) form*
  - is called the **characteristic function** (or **eigenfunction**) of the system
- This means that an everlasting sinusoid ($s = \pm j\omega$) is also a characteristic function of the system

Response of an LTI System to e^{st}

- An LTI system's response to an everlasting exponential e^{st}:

$$y(t) = h(t) * e^{st} = \int_{-\infty}^{\infty} h(\tau)e^{s(t-\tau)}d\tau = e^{st}\int_{-\infty}^{\infty} h(\tau)e^{-s\tau}d\tau$$

- The integral on the right side: a function of the complex variable *s*, denoted $H(s)$, thus:

$$y(t) = H(s)e^{st}$$

- Note: for a given *s* , $H(s)$ = a constant
  - i.e. the input and output are the same (with a multiplicative constant)
- $H(s)$ is called the **transfer function** of the system
  - describes the **input / output relationship**

Frequency Response of LTI Systems

- Apply an input e^{st} , setting $s = j\omega$

$$e^{j\omega t} \Rightarrow H(j\omega)e^{j\omega t}$$

- Note: $\cos \omega t$ is the real part of $e^{j\omega t}$

$$\cos \omega t \Rightarrow \mathrm{Re}\{H(j\omega)e^{j\omega t}\}$$

- Expressing $H(j\omega)$ in polar form

$$\cos \omega t \Rightarrow |H(j\omega)| \cos[\omega t + \angle H(j\omega)]$$

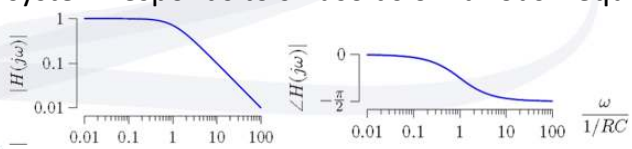
Amplitude response

Phase response

- ∴ LTI response to sinusoid: a **sinusoid of the same frequency**
  - amplitude scaled by $|H(j\omega)|$
  - phase shifted by $\angle H(j\omega)$
- $H(j\omega)$ is termed the **frequency response** of the system (special case of $H(s)$)

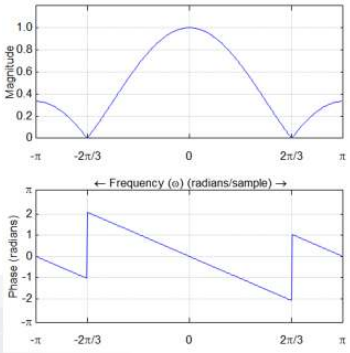
Frequency Response of LTI Systems

- E.g. If $|H(j10)| = 3$ and $\angle H(j10) = -30°$ for a certain system
  - For input: $x(t) = 5\cos(10t + 50°)$
  - Output: $y(t) = 15\cos(10t + 20°)$
- $|H(j\omega)|$ is the amplitude **gain** of the system
- Plot of $|H(j\omega)|$ against frequency : **amplitude** (or **magnitude**) **response**
- Plot of $\angle H(j\omega)$ against frequency: the **phase response**
- Plots show how the system responds to sinusoids of various frequencies

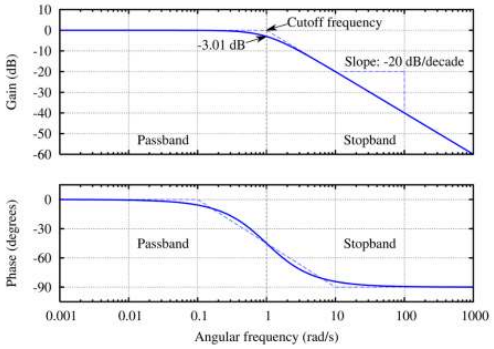


Frequency Response plots

- Frequency response plots
- Can use linear scales



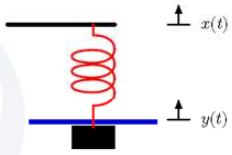
- More commonly, use log scales
  - called Bode plots



Frequency Response Example

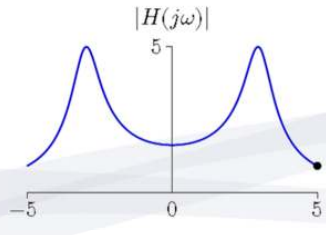
- Consider a mass – spring – dashpot system
- It can be shown that the frequency response of this system is:

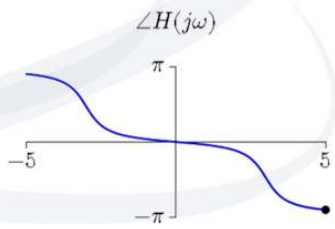
$$H(j\omega) = \frac{K}{-\omega^2 M + j\omega B + K}$$



$$F = Ma = M\ddot{y}(t) = K(x(t) - y(t)) - B\dot{y}(t)$$

$$M\ddot{y}(t) + B\dot{y}(t) + Ky(t) = Kx(t)$$

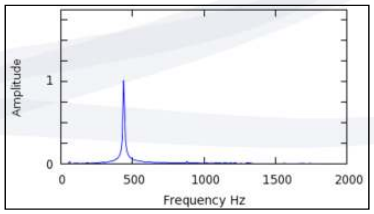




Frequency Spectrum



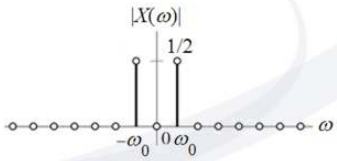
- To make use of the frequency response
- Need to know the frequency content of our signal: the **frequency spectrum**
- Tuning fork,
  - generates a signal with a single frequency component
  - $x(t) = \sin(\omega_0 t)$
    - i.e. a sinusoid with a frequency of ω_0
    - amplitude of 1

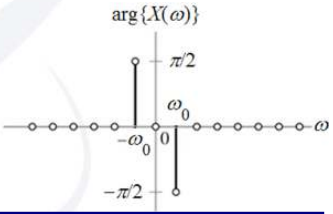


$$m\frac{d^2x(t)}{dt^2} + kx(t) = 0;\ x(0) = 0;\ \dot{x}(0) = \omega_0$$
$$\therefore mr^2 + k = 0 \Rightarrow r_{1,2} = \pm j\sqrt{k/m} = \pm j\omega_0$$
$$\therefore x(t) = c_1 e^{j\omega_0 t} + c_2 e^{-j\omega_0 t}$$
$$\begin{cases} x(0) = c_1 + c_2 = 0 \\ \dot{x}(0) = j\omega_0(c_1 - c_2) = \omega_0 \end{cases} \Rightarrow \begin{cases} c_1 = \frac{1}{2j} \\ c_2 = -\frac{1}{2j} \end{cases}$$
$$\therefore x(t) = \frac{e^{j\omega_0 t} - e^{-j\omega_0 t}}{2j} \Rightarrow x(t) = \sin(\omega_0 t)$$

Frequency Spectrum Plot

- **Spectral plot:** each frequency component is represented as a single **spectral line** at the appropriate frequency
- A two-sided spectral plot
  - both positive and negative frequency terms from complex exponential terms
    - Amplitude of negative frequency component = complex conjugate of the corresponding positive frequency component
    - a general property of the two-sided spectra of real signals

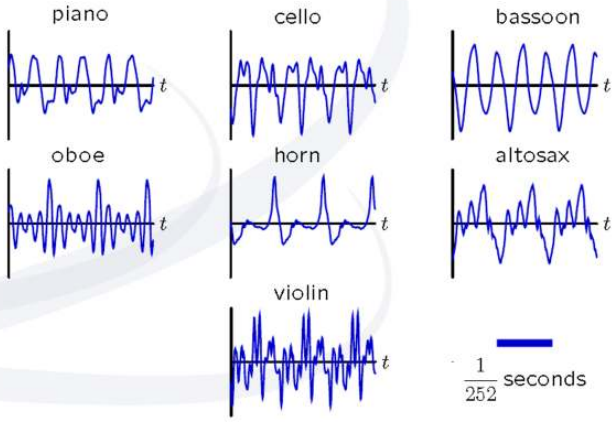




$$x(t) = \frac{1}{2j}e^{j\omega_0 t} - \frac{1}{2j}e^{-j\omega_0 t} = \frac{1}{2}e^{-j\pi/2}e^{j\omega_0 t} + \frac{1}{2}e^{j\pi/2}e^{-j\omega_0 t}$$

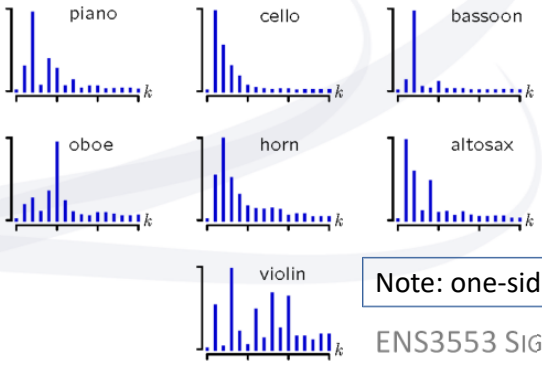
Frequency Spectrum Plot

- Most signals have more than one spectral component
- E.g. sounds produced by musical instruments
  - not "pure" tones like the sound produced by the tuning fork



Frequency Spectrum Plot

- These instruments produce sounds that consist of
  - a ***fundamental*** sinusoidal component at the frequency of the note being played
  - ***harmonics*** at multiples of that frequency
- Having distinct frequency components is characteristic of periodic signals



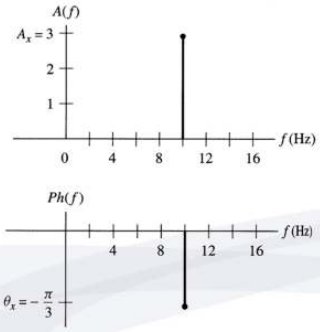
Note: one-sided spectral plots

Frequency Spectrum Plot

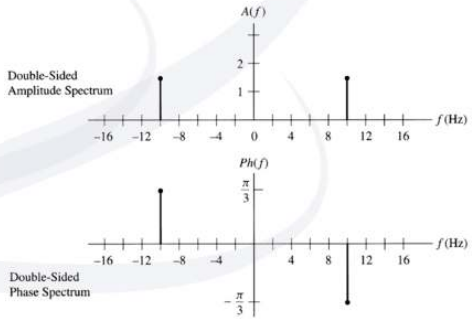
$$\cos x = \frac{e^{jx} + e^{-jx}}{2}$$

- Consider a 10Hz (20π rad/s) sinusoidal signal:
  - $x(t) = 3\cos(20\pi t - \pi/3) = \left(1.5e^{-j\pi/3}\right)e^{j20\pi t} + \left(1.5e^{j\pi/3}\right)e^{-j20\pi t}$

Single sided spectra



Two-sided spectra

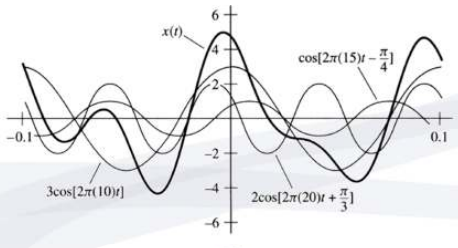


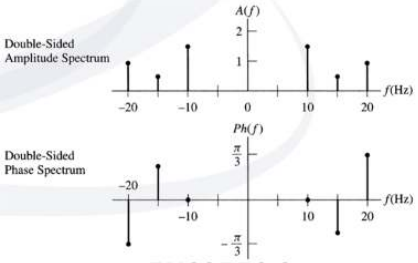
Frequency Spectrum Plot

- Consider a composite sinusoidal signal:

  $$x(t) = 3\cos(20\pi t) + \cos(30\pi t - \pi/4) + 2\cos(40\pi t + \pi/3)$$

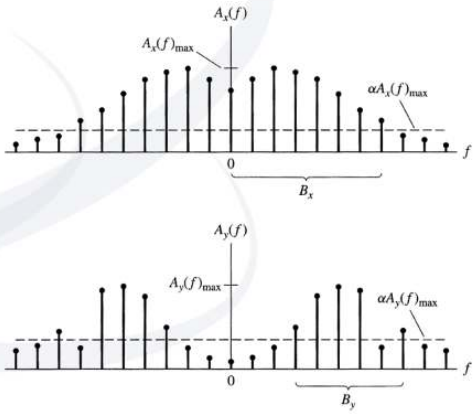
- Three components with frequencies of 10Hz, 15Hz and 20Hz
- Composite signal is periodic with a period of 0.2s
  - the lowest common multiple of the three periods





Signal Bandwidth

- **Bandwidth of a signal**: frequency band over which the frequency components of the signal exceed a certain amplitude threshold
- Threshold: defined as a fraction of the maximum amplitude of the signal
  - $\alpha A_{max},\ \alpha < 1$
- Commonly the half-power point is used
  - i.e. 0.707 x the maximum signal amplitude
  - Also known as -3dB point
- Other thresholds can also be used



Application: Audio Compression

- Standard CD quality audio:
  - sampled 44,100 times per second
  - each sample is quantised to 16 bits (2 bytes)
  - separate samples for the left and right channels (for stereo playback)
- Number of bits to be stored for each second of audio recording
  - 44,100 samples/second * 16 bits/sample * 2 channels = 1,411,200 bits per sec
  - = 176,000 bytes per sec
- So average length musical track of 3 minutes = about 32 Mb of space
- Need to compress the signal
  - but without negatively impacting the audio quality

Application: Audio Compression

- The MP3 format
  - a compression system for music
  - exploits the system characteristics of the human ear
    - human hearing is limited to the range between about 20Hz and 20kHz
    - human ear hear certain frequencies much better than others
      - in particular the frequency band from 2kHz to 5kHz
    - if there are two sounds of similar frequency playing simultaneously, we hear the louder one but cannot hear the softer (masking)
    - the effective "resolution" of human hearing is no better than 2Hz

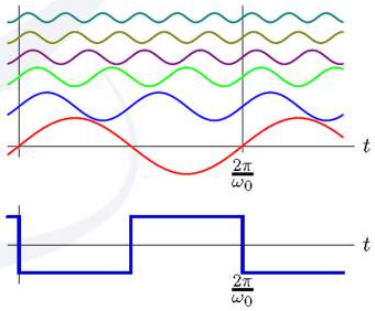
Application: Audio Compression

- Basic principle for MP3 compression:
  - Find frequency content (frame by frame)
  - Exclude components that people can't perceive and allocate bits to the rest according to their relative importance
  - Compress bit stream (Huffman coding)
- End up with is a "near-CD-quality" song
  - compressed by a factor of generally 10 to 14
    - the average song compresses down to about 3 Mb
  - much more efficient storage or transfer

Harmonic Representations



- Is it possible to represent ALL periodic signals with harmonics?
- What about discontinuous signals?
- Fourier claimed YES
  - even though all harmonics are continuous!
  - the idea that a discontinuous signal could be written as a sum of continuous signals was ridiculed by some (e.g. Lagrange)
- Assume the answer is YES and see if the answer makes sense ...



The Fourier Series

- A periodic signal $x(t)$ with period T_0:

$$x(t) = x(t + T_0) \text{ for all } t$$

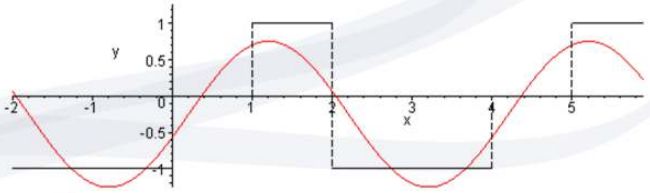
  - **Fundamental period**: smallest value of T_0 that satisfies above condition
- Consider a signal $x(t)$
  - made up of sines and cosines of the same frequency, ω_0, and all its harmonics
  - with arbitrary amplitudes
    - including the 0th, i.e. constant (or DC)

$$x(t) = a_0 + \sum_{n=1}^{\infty} a_n \cos n\omega_0 t + b_n \sin n\omega_0 t$$

- Can be proven: sum is a periodic signal with the same period as the fundamental regardless of the values of a_n and b_n

The Fourier Series

- Theoretically can construct a variety of periodic signals by changing the values of a_n and b_n
- Fourier showed can construct almost ***any*** periodic signal
  - by appropriately choosing coefficient values for the infinite series
  - the **Fourier coefficients**
  - Series in this form: the **trigonometric Fourier series**



The Fourier Series

- For the trigonometric Fourier series the Fourier coefficients are given by:

$$a_0 = \frac{1}{T_0}\int_{T_0} x(t)dt \qquad \left(T_0 = \frac{1}{f_0} = \frac{2\pi}{\omega_0}\right)$$

$$a_n = \frac{2}{T_0}\int_{T_0} x(t)\cos n\omega_0 t\, dt$$

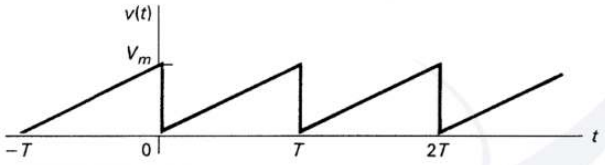
$$b_n = \frac{2}{T_0}\int_{T_0} x(t)\sin n\omega_0 t\, dt$$

n = 0, DC component
n = 1, fundamental frequency
n = 2, 2nd harmonic
…

Integration over one cycle, i.e. $t_0 \rightarrow t_0 + T_0$

Example: Trigonometric Fourier Series

- Find the trigonometric Fourier series of the following waveform:



$$v(t) = \frac{V_m}{T}t$$

- a_0 is the average of $v(t)$ over one cycle
  - For a voltage signal like = dc voltage of the signal

$$a_0 = \frac{1}{T}\int_0^T v(t)dt = \frac{1}{T}\int_0^T \frac{V_m}{T}t\,dt = \frac{V_m}{2}$$

Example: Trigonometric Fourier Series

$$a_n = \frac{2}{T}\int_0^T v(t)\cos n\omega_0 t\,dt = \frac{2}{T}\int_0^T \frac{V_m}{T}t\cos n\omega_0 t\,dt$$

$$= \frac{2V_m}{T^2}\left[\frac{1}{n^2{\omega_0}^2}\cos n\omega_0 t + \frac{t}{n\omega_0}\sin n\omega_0 t\right]\Bigg|_0^T$$

$$= \frac{2V_m}{T^2}\left(\frac{1}{n^2{\omega_0}^2}(\cos 2\pi n - 1)\right) = 0 \text{ for all } n$$

Example: Trigonometric Fourier Series

$$b_n = \frac{2}{T}\int_0^T v(t)\sin n\omega_0 t\,dt = \frac{2}{T}\int_0^T \frac{V_m}{T}t\sin n\omega_0 t\,dt$$

$$= \frac{2V_m}{T^2}\left[\frac{1}{n^2{\omega_0}^2}\sin n\omega_0 t - \frac{t}{n\omega_0}\cos n\omega_0 t\right]\Bigg|_0^T$$

$$= \frac{2V_m}{T^2}\left(0 - \frac{T}{n\omega_0}\cos 2\pi n\right) = -\frac{V_m}{\pi n}$$

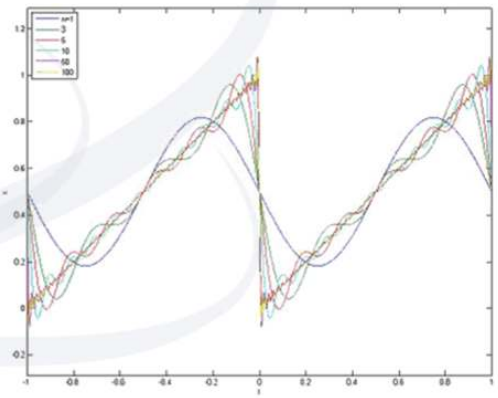
Example: Trigonometric Fourier Series

- The Fourier series for $v(t)$ is therefore:

$$v(t) = \frac{V_m}{2} - \frac{V_m}{\pi}\sum_{n=1}^{\infty}\frac{1}{n}\sin n\omega_0 t$$

$$= \frac{V_m}{2} - \frac{V_m}{\pi}\sin\omega_0 t - \frac{V_m}{2\pi}\sin 2\omega_0 t$$

$$-\frac{V_m}{3\pi}\sin 3\omega_0 t - \cdots$$



A Word on Symmetry

- Cosine functions: **even** symmetry
- Sine functions: **odd** symmetry
- If a signal has even symmetry
  - will contain only cosine terms
  - all $b_n = 0$
- if signal has odd symmetry
  - will contain only sine terms
  - all $a_n = 0$
- Note: due to the symmetry (odd or even) only need to know the signal waveform over half a period to reconstruct the full waveform
  - can compute the Fourier coefficients by integrating over only half the period

Compact Form of Fourier Series

- If $x(t)$ is real, then the coefficients a_n and b_n are real for all n
- Trigonometric Fourier series can be expressed in a **compact form**:

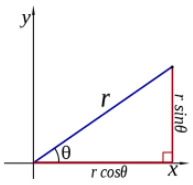
$$x(t) = C_0 + \sum_{n=1}^{\infty} C_n \cos(n\omega_0 t + \theta_n)$$

where:

$$C_0 = a_0$$

$$C_n = \sqrt{{a_n}^2 + {b_n}^2} \quad \text{(magnitude of } n^{\text{th}} \text{ harmonic)}$$

$$\theta_n = \tan^{-1}\left(\frac{-b_n}{a_n}\right) \quad \text{(phase of } n^{\text{th}} \text{ harmonic)}$$



Also: $\sin(\theta) = \cos\left(\theta - \frac{\pi}{2}\right)$

Exponential Fourier Series

- Using Euler's equality, can express the Fourier series in more compact form
  - *see textbook for derivation*

where

$$x(t) = \sum_{n=-\infty}^{\infty} D_n e^{jn\omega_0 t}$$

$$D_n = \frac{1}{T_0}\int_{T_0} x(t)e^{-jn\omega_0 t}dt$$

Note:

$$D_0 = a_0$$

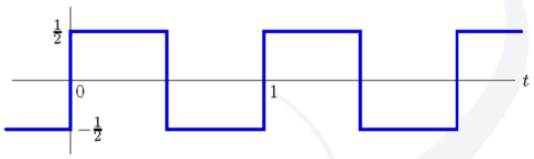
$$D_n = \frac{a_n - jb_n}{2}$$

$$D_{-n} = D_n^*$$

- Generally the preferred form, due to
  - compactness of the equations
  - ease of mathematical manipulation
- **Note:** gives rise to the two-sided frequency spectra

Question 1

Let a_n, b_n and D_n represent the Fourier series coefficients of the following square wave:

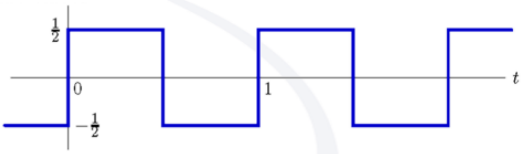


Which of the following statements are true:

1. $a_n = 0$
2. D_n is real valued
3. There are an infinite number of non-zero D_n
4. All of the above

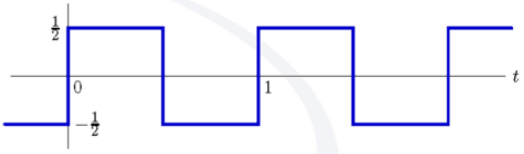
Web Address: http://pollev.com/signals

Q1 Solution



- Waveform is odd
- Therefore $a_n = 0$, for all n
- $D_n = \frac{a_n - jb_n}{2}$ Since $a_n = 0$, $D_n = j\frac{b_n}{2}$ i.e. **complex**

Q1 Solution

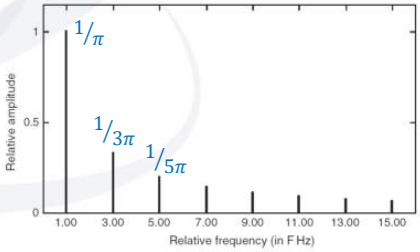


- $D_n = \frac{1}{T_0}\int_{T_0} x(t)e^{-jn\omega_0 t}dt$

$$D_n = \frac{1}{T_0}\int_{T_0} x(t)e^{-j\frac{2\pi}{T}nt}dt = -\frac{1}{2}\int_{-\frac{1}{2}}^{0} e^{-j2\pi nt}dt + \frac{1}{2}\int_{0}^{\frac{1}{2}} e^{-j2\pi nt}dt$$

$$= \frac{1}{j4\pi n}\left(2 - e^{-j\pi n} - e^{-j\pi n}\right)$$

$$= \begin{cases} \frac{1}{j\pi n}; & \text{if } n \text{ is odd} \\ 0; & \text{otherwise} \end{cases}$$



Effect of Differentiation and Integration

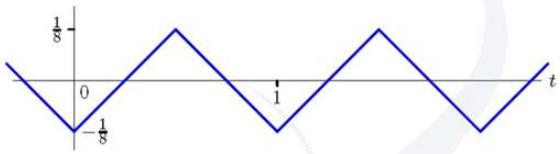
- Differentiating a periodic signal represented by an exponential Fourier series:

$$\frac{dx(t)}{dt} = \frac{d}{dt}\sum_{n=-\infty}^{\infty} D_n e^{jn\omega_0 t} = \sum_{n=-\infty}^{\infty} jn\omega_0 D_n e^{jn\omega_0 t}$$

- Effect of differentiating a periodic signal
  - to multiply the original Fourier coefficients by $jn\omega_0$
  - i.e. differentiation increases the relative weight of the higher harmonics
- The effect of integrating a periodic signal is the opposite
  - the original Fourier coefficients are divided by $jn\omega_0$

Question 2

Let D_n represent the Fourier series coefficients of the following triangular wave:



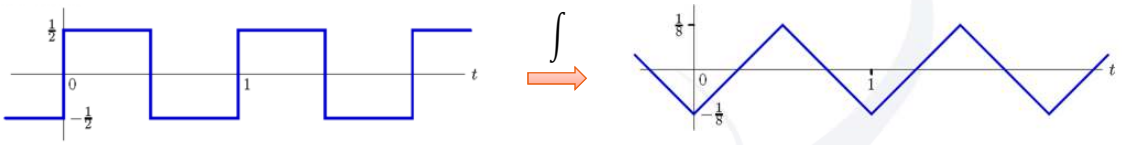
Select all of the following statements that are true:

1. $D_n = 0$ if n is even
2. D_n is real valued
3. $|D_n|$ decreases with n^2
4. There are an infinite number of non-zero D_n

Web Address: http://pollev.com/signals

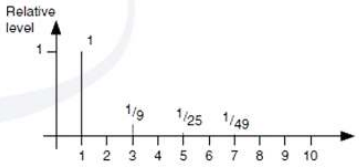
Q2 Solution

- The triangle wave is the integral of the square wave:



- Therefore, the Fourier coefficients of the triangle waveform are $\frac{1}{jn\omega_0} = \frac{T}{j2\pi n} = \frac{1}{j2\pi n}$ times those of the square wave:

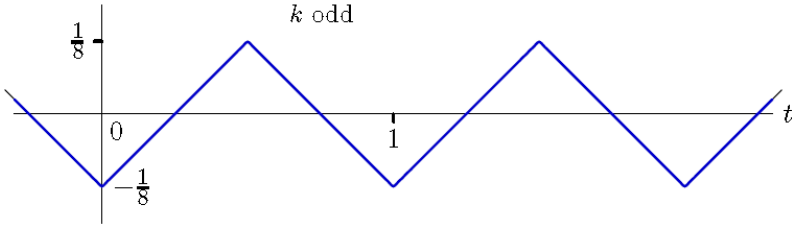
$$D_n = \frac{1}{j2\pi n} \times \frac{1}{j\pi n} = -\frac{1}{2\pi^2 n^2}; \qquad n \text{ odd}$$



Convergence of the Fourier Series

- We can visualise the convergence of the Fourier series by incrementally adding terms
- Example: triangle wave

$$\sum_{\substack{k=-39 \\ k \text{ odd}}}^{39} \frac{-1}{2k^2\pi^2} e^{j2\pi kt}$$

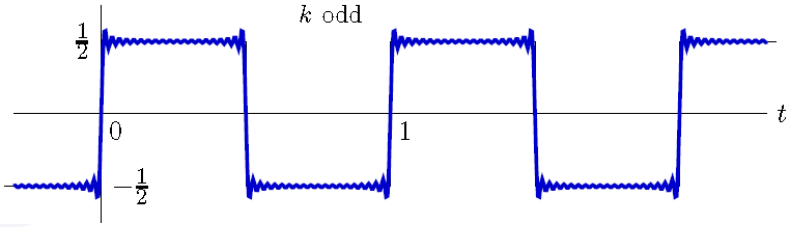


Convergence of the Fourier Series

- Can visualise the convergence of the Fourier series by incrementally adding terms
- Example: square wave

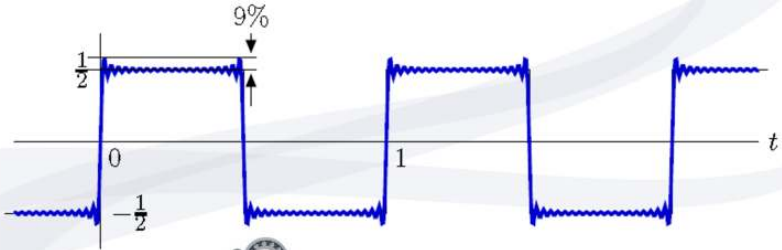
Notice anything odd about the convergence in this case?

$$\sum_{\substack{k=-39 \\ k \text{ odd}}}^{39} \frac{1}{jk\pi} e^{j2\pi kt}$$



Convergence of the Fourier Series

- When there is a jump discontinuity in $x(t)$
  - see "ringing" at the point of discontinuity
  - called the **Gibbs phenomenon**
- Expect the series to converge exactly to $x(t)$ as $n \to \infty$,
  - yet even for large n, the truncated series exhibits an overshoot approaching a value of 9% near the jump



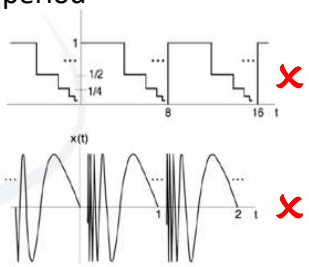
Note: 9% is more or less constant irrespective of the value of n

Convergence of the Fourier Series

- Frequency of oscillation of the components of the synthesized signal is nf_0
  - the width of the overshoot spike is $\sim 1/2nf_0$
  - thus as n increases the spike width diminishes
  - as $n \to \infty$ the width $\to 0$
- In terms of the *error energy*
  - i.e. difference in signal energy between $x(t)$ and Fourier over one period
  - as $n \to \infty$ the error energy $\to 0$
- While the series may differ from $x(t)$ at some points
  - the error signal power tends to zero for increasing n
    - the series converges in the mean at the points of discontinuity

Convergence of the Fourier Series

- To guarantee that the Fourier series converges
  - pointwise at all points where $x(t)$ is continuous
  - to a value mid-way between the values of $x(t)$ on either side of discontinuities
- Some conditions must be met – the **Dirichlet conditions**:
  - The function $x(t)$ must be absolutely integrable over one period
    - i.e. $\int_T |x(t)|dt < \infty$
  - The function $x(t)$ must have a finite number of finite discontinuities in one period
  - The function $x(t)$ must contain a finite number of maxima and minima in one period



Parseval's Theorem

- The **average power of a periodic signal** $x(t)$
- Can be calculated from the time domain representation of the signal as:

$$P = \frac{1}{T}\int_T x^2(t)dt$$

- Can be calculated from the frequency domain representation of the signal as:

$$P = \sum_{n=-\infty}^{\infty} |D_n|^2 = {a_0}^2 + \frac{1}{2}\sum_{n=1}^{\infty}\left({a_n}^2 + {b_n}^2\right)$$

Sum of the powers of the Fourier components

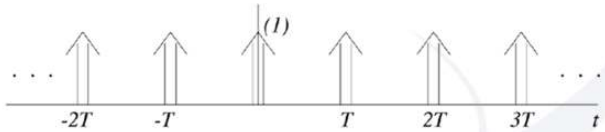
- This is a form of **Parseval's Theorem** as applied to power signals

Power of sinusoid = $\frac{C^2}{2}$ irrespective of frequency or phase

- Refer Tutorial 2

Another Example: Periodic Impulse Train

- Consider the Fourier series of a periodic impulse train
  - important for sampling *(more on this later)*



$$D_n = \frac{1}{T}\int_{-T/2}^{T/2} x(t)e^{-jn\omega_0 t}dt = \frac{1}{T}\int_{-T/2}^{T/2} \delta(t)e^{-jn\omega_0 t}dt$$

$$= \frac{1}{T} \quad \text{for all } n!$$

- All components have the same frequency and the same phase

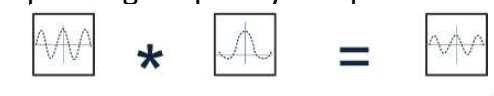
LTI System Response to Periodic Signals

- Can use the Fourier series to easily calculate the system response to a periodic input
  - due to the superposition property of LTI systems
- For an LTI system with a frequency response given by $H(j\omega)$
  - the output $y(t)$ in response to a periodic input $x(t)$
    = periodic signal with Fourier series coefficients related to those of the input
    - adjusted according to the frequency response of the system at each harmonic

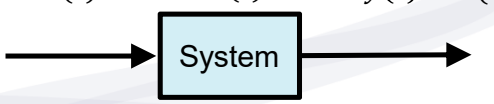
$$y(t) = \sum_{n=-\infty}^{\infty} D_{ny} e^{jn\omega_0 t}, \qquad \text{where } D_{ny} = D_{nx} H(jn\omega_0)$$

LTI System Response to Periodic Signals

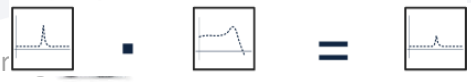
- If input signal can be represented by its frequency spectrum $X(j\omega)$
- And system frequency response, $H(j\omega)$, is known
- Then frequency spectrum of the output $Y(j\omega)$ is obtained by multiplying each corresponding frequency component of $X(j\omega)$ and $H(j\omega)$



$x(t)$ $\quad$ $h(t)$ $\quad$ $y(t) = x(t) * h(t)$

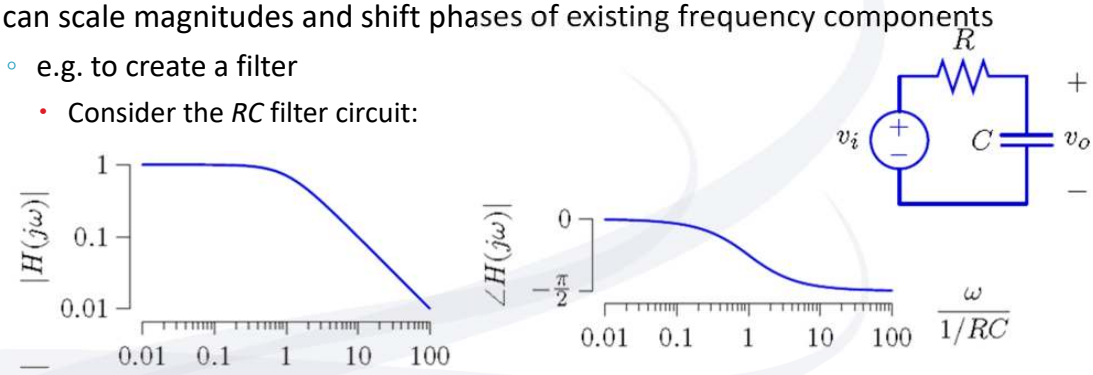


$X(j\omega)$ $\quad$ $H(j\omega)$ $\quad$ $Y(j\omega) = X(j\omega)H(j\omega)$



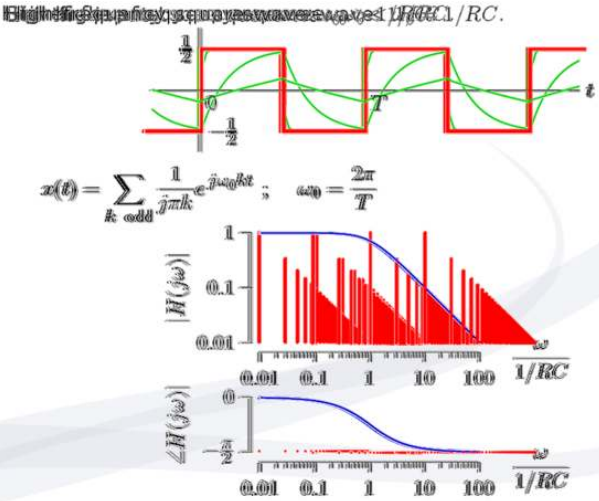
Application: Filtering

- LTI systems cannot create new frequencies
- can scale magnitudes and shift phases of existing frequency components
  - e.g. to create a filter
    - Consider the *RC* filter circuit:



- What type of filter is this?

Application: Filtering

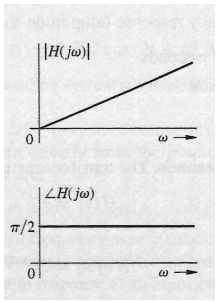


Graphical Representation

- Frequency response plot - tells us at a glance what the system does to sinusoids of different frequencies
- 2 important points about the frequency response of an LTI system:
  - The response usually varies with frequency
    - sinusoids of different frequencies are treated differently by the system
  - By appropriate choice of the system parameters, a wide variety of frequency response shapes can be realised



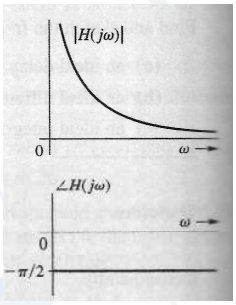
Ideal Differentiator and Integrator response



- Ideal differentiator

$$H(j\omega) = j\omega = \omega e^{j\pi/2}$$

  - Amplifies higher frequency components
    - Including noise!



- Ideal integrator

$$H(j\omega) = \frac{1}{j\omega} = \frac{-j}{\omega} = \frac{1}{\omega} e^{-j\pi/2}$$

  - Suppresses higher frequency components

Frequency response of ideal delay

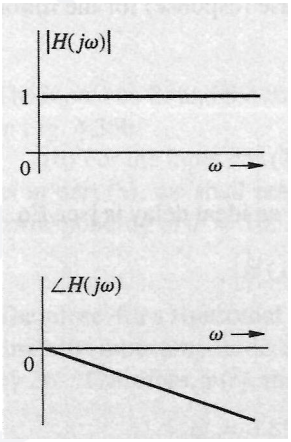
- For ideal delay of T seconds

  $H(s) = e^{-sT}$ (time shifting property)

- Therefore

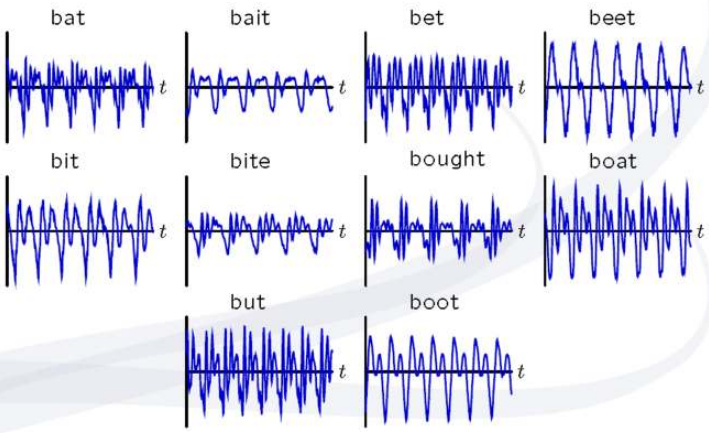
  $|H(j\omega)| = \left|e^{-j\omega T}\right| = 1$ and $\angle H(j\omega) = -\omega T$

- Delay of *T*
  - has no effect on amplitude
  - Linear phase shift with a gradient –*T*



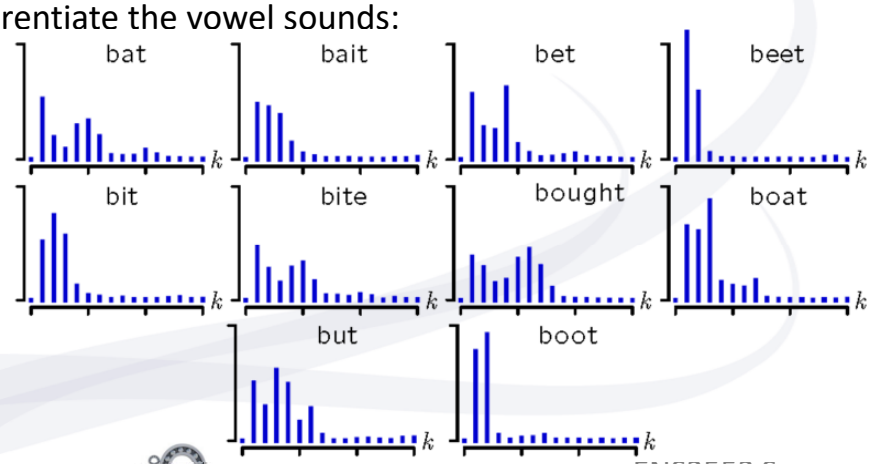
Application: Speech Modelling

- Vowel sounds can be modelled as quasi periodic signals:



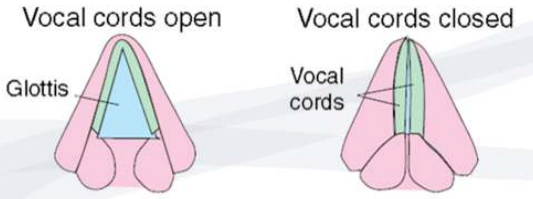
Application: Speech Modelling

- The harmonic content then becomes a natural way to describe and differentiate the vowel sounds:

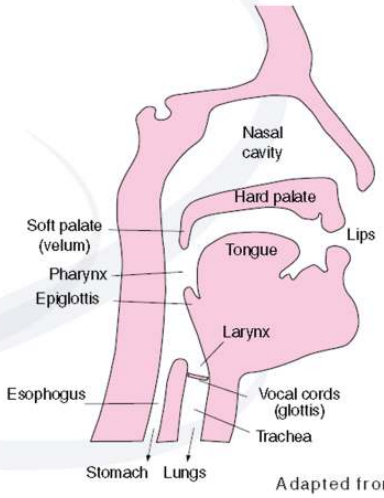


Application: Speech Modelling

- **Speech**: generated by the passage of air from the lungs, through the vocal cords, mouth, and nasal cavity
- Controlled by complicated muscles, the vocal cords are set into vibrational motion by passage of air from the lungs



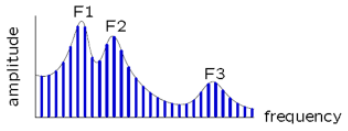
Adapted from T.F. Weiss



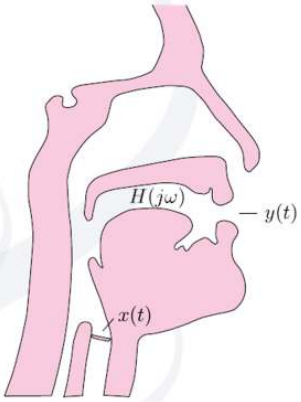
Adapted from T.F. Weiss

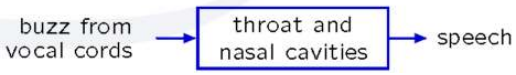
Application: Speech Modelling

- Vibrations of the vocal cords are "filtered" by the mouth and nasal cavities to generate speech
- Resonant frequencies of the vocal tract:



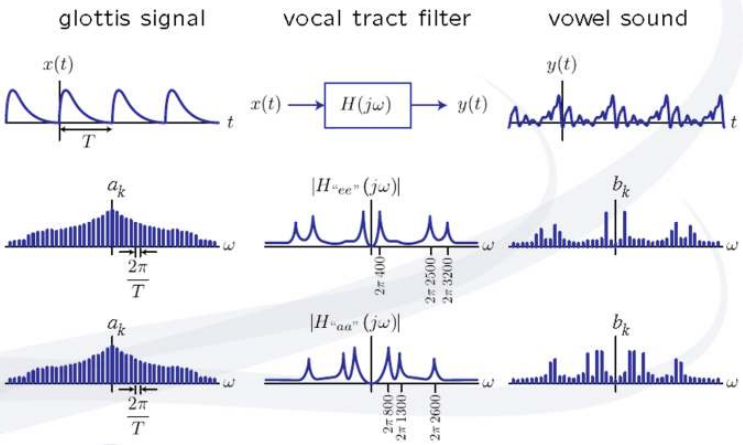
	Formant	heed	head	had	hod	haw'd	who'd
Men	F1	270	530	660	730	570	300
	F2	2290	1840	1720	1090	840	870
	F3	3010	2480	2410	2440	2410	2240
Women	F1	310	610	860	850	590	370
	F2	2790	2330	2050	1220	920	950
	F3	3310	2990	2850	2810	2710	2670
Children	F1	370	690	1010	1030	680	430
	F2	3200	2610	2320	1370	1060	1170
	F3	3730	3570	3320	3170	3180	3260





Application: Speech Modelling

- Same glottis signal + different formants → different vowels:

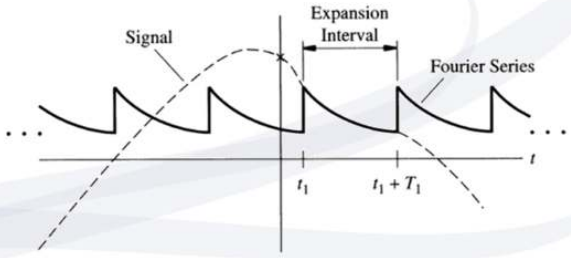


Limitations of Fourier Analysis

- Fourier series : periodic signals as a weighted sum of everlasting exponentials
  - valuable for many applications
- Some limitations with respect to the analysis of linear systems:
  - The Fourier series can only be used for periodic inputs
    - practical inputs are aperiodic *(solution covered shortly)*
  - The Fourier methods can be applied to BIBO (or asymptotically) stable systems, but it cannot handle unstable or marginally stable systems
- Second limitation can be overcome by allowing *s* to take on complex values
  - rather than being limited to $j\omega$
  - leads to the **Laplace integral** *(to be covered in a couple of weeks)*

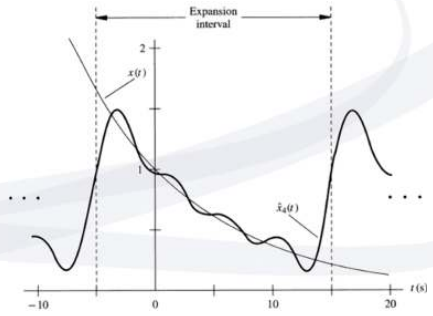
Application to Aperiodic Signals

- The Fourier series can be used to model an aperiodic signal over a fixed interval
- If the aperiodic signal is $x(t)$ then the shape of $x(t)$ over the expansion interval is repeated to create a periodic signal:



Application to Aperiodic Signals

- These Fourier series models usually truncated to a finite number of terms
  - e.g. to model an aperiodic function over a fixed interval
  - include only the harmonics up to and including the fourth harmonic
  - end up with an approximation function:



Next week we shall look at an alternate method to represent aperiodic signals

Summary

- How to represent a very large class of periodic signals as an infinite sum of sinusoids (or exponentials) via the Fourier series
- The Fourier series represents a weighted sum of a sinusoid of frequency, f_0 (the fundamental frequency of the original signal) and its harmonics
- Can plot the amplitude and phase of these sinusoids against frequency to produce a spectral plot
  - provides significant insight into the frequency domain properties of the signal
- Can predict the output from a system with knowledge of the spectral content of an input signal and the frequency response of the system